Def. Let X, Y be Metric Spaces, and $f_n : X \to Y$ be function defined for each $n \in \mathbb{N}$.
For each $x \in X$, define $f(x) \overset{\text{def}}{=} \lim_{n\to\infty} f_n(x)$ if the limit exists.

This f is called limit function of $\{f_n\}$.

Def. Suppose that a sequence of functions $\{f_n\}$ on X into Y satisfies that;
the limit function f is defined on X and

For all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow d_Y(f_n(x), f(x)) \leq \varepsilon$, $\forall x \in E$.

Then, we say that $\{f_n\}$ converges uniformly on X to f. Denote $f_n \rightrightarrows f$.
Note: $\forall x \in X$, $d_Y(f_n(x), f(x)) \leq \varepsilon \iff \sup_{x\in X} d_Y(f_n(x), f(x)) \leq \varepsilon$. That is,

$$f_n \rightrightarrows f \text{ on } X \iff \lim_{n\to\infty} \sup_{x\in X} d_Y(f_n(x), f(x)) = 0.$$

Thm. Let $\{f_n\}$ be functions defined on any Metric Space X into Complete Metric Space Y. Then,

In General Metric Space

$\{f_n\}$ converges uniformly on X to f if and only if $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t $m, n \geq N \Rightarrow \sup_{x\in X} d_Y(f_n(x), f_m(x)) \leq \varepsilon$

In Complete Metric space

Proof. If $f_n \rightrightarrows f$, then,

$$\sup_{x\in X} d_Y(f_n(x), f_m(x)) \leq \sup_{x\in X} d_Y(f_n(x), f(x)) + \sup_{x\in X} d_Y(f(x), f_m(x)) \to 0$$

give the result. (간단해 보이지만, Supremum 의 성질과 triangle inequality 가 합쳐진 좋은 예시임.)
Conversely, since Cauchy sequence in the Complete Metric Space converges, the limit function f exist.
Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ such that the Cauchy condition holds.
For fixed $n \geq N$, for all $x \in X$ and $m \geq N$, $d_Y(f_n(x), f_m(x)) \leq \varepsilon$.
So, letting $m \to \infty$, we obtain $d_Y(f_n(x), f(x)) \leq \varepsilon$, for all $x \in X$. Thus proved. ∎

Fail Try.

$$\lim_{m\to\infty} \sup_{x\in X} d_Y(f_m(x), f_n(x)) = \sup_{x\in X} d_Y(f(x), f_n(x))$$

gives the result. (More specifically, let $n \in \mathbb{N}$ fixed. Observe that

$$\left| \sup_{x\in X} d_Y(f_m(x), f_n(x)) - \sup_{x\in X} d_Y(f(x), f_n(x)) \right|$$

$$\sup_{x\in X} d_Y(f(x), f_n(x)) = \sup_{x\in X} d_Y\left(\lim_{m\to\infty} f_m(x), f_n(x)\right) \overset{d_Y \text{ Continuous}}{=} \sup_{x\in X} \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) \overset{(*)}{=} \lim_{m\to\infty} \sup_{x\in X} d_Y(f_m(x), f_n(x))$$

$$\left| \sup_{x\in X} \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) - \sup_{x\in X} d_Y(f_m(x), f_n(x)) \right| \leq \sup_{x\in X} \left| \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) - d_Y(f_m(x), f_n(x)) \right| \to 0$$

gives $(*)$.

「Uniform Convergence and Continuity.

「Thm. Suppose that $f_n, f : X \to Y$ are functions where X is Metric, Y is Complete Metric.sp.
Let $x \in X$ be a limit point of X.
If for each $n \in \mathbb{N}$, the limit $\lim_{t \to x} f_n(t)$ exists and $f_n \to f$ uniformly on X,
then

$$\lim_{t \to x} \lim_{n \to \infty} f_n(t) = \lim_{n \to \infty} \lim_{t \to x} f_n(t)$$

Proof. For each $k \in \mathbb{N}$, take $A_k = \lim_{t \to x} f_k(t)$. And, Since the uniformly Convergence,
given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ Such that

$$m, n \geq N \Rightarrow d_Y(f_n(t), f_m(t)) \leq \varepsilon \text{ for all } t \in X.$$

And, for such $m, n \geq N$, let $x_0 \in X$ such that

$$d_Y(A_m, f_m(x_0)) < \varepsilon \quad \text{and} \quad d_Y(A_n, f_n(x_0)) < \varepsilon$$

(for some δ_i, $\delta > 0$, $t \in B_\delta(x) \Rightarrow d_Y(f_k(t), A_k) < \varepsilon$, by def. of A_k)
Now,

$$d_Y(A_m, A_n) \leq d_Y(A_m, f_m(x_0)) + d_Y(f_m(x_0), f_n(x_0)) + d_Y(f_n(x_0), A_n) \leq 3\varepsilon,$$

Therefore $\{A_k\}$ is a Cauchy sequece in Y, So it is Convergent.
Let $A = \lim_{n \to \infty} A_n$ Then,

$$d_Y(f(t), A) \leq \underbrace{d_Y(f(t), f_n(t))}_{f_n \to f \text{ uniformly}} + \underbrace{d_Y(f_n(t), A_n)}_{\substack{f_n(t) \to A_n \\ \text{as } t \to x}} + \underbrace{d_Y(A_n, A)}_{A_n \to A} \to 0$$

Rigorously, given $\varepsilon > 0$, let $N_1, N_2 \in \mathbb{N}$ s.t

$$n \geq N_1 \Rightarrow d_Y(A_n, A) < \varepsilon \quad \text{and} \quad n \geq N_2 \Rightarrow \forall t \in X,\ d_Y(f_n(t), f(t)) < \varepsilon.$$

Now, take $n \geq \max\{N_1, N_2\}$ and $\delta > 0$ s.t

$$t \in B_\delta(x) \Rightarrow d_Y(f_n(t), A_n) < \varepsilon. \quad \blacksquare$$

「Corollary. If $\{f_n : X \to Y\}$ is a sequence of Continuous maps on a Metric space X into Complete Y,
and $f_n \rightrightarrows f$, then f is Continuous.

「Uniform Convergence with Continuity

「Thm. Suppose that f_n $(n\in\mathbb{N})$ are functions on a Metric X into Complete Metric Y,
Let $x\in X$ be a limit point of X.
If for each $n\in\mathbb{N}$, the limit $\lim_{t\to x} f_n(t)$ exists and f_n Converges uniformly on X,
then $$\lim_{t\to x}\lim_{n\to\infty} f_n(t) = \lim_{n\to\infty}\lim_{t\to x} f_n(t).$$

Proof. By the Uniformly Convergence, given $\varepsilon>0$, there exists $N\in\mathbb{N}$ s.t.
$$n,m\geq N \Rightarrow d_Y\left(f_m(t), f_n(t)\right)\leq\varepsilon \text{ for any } t\in X.$$

「Thm. Let K be a Compact space, and
$f_n : K\to\mathbb{R}$ be continuous functions which converges pointwise to a Continuous function f on K.
If $f_n\geq f_{n+1}$ for all $n\in\mathbb{N}$, then $f_n\to f$ uniformly.

Proof. Put $g_n = f_n - f$. Then, g_n is Continuous and $g_n\to 0$ pointwise, and $g_n\geq g_{n+1}$.
Given $\varepsilon>0$, let $K_n = \{x\in K \mid g_n(x)\geq\varepsilon\} = g_n^{-1}\left[[\varepsilon,\infty)\right]$
Since g_n is Continuous, K_n is a closed set in a Compact space, so it is Compact.
And, Since $g_n\geq g_{n+1}$, $K_n\supseteq K_{n+1}$ (if $g_{n+1}(x)\geq\varepsilon$, then $g_n(x)\geq g_{n+1}(x)\geq\varepsilon$)
Fix $x\in X$. Since $g_n(x)\to 0$ as $n\to\infty$, for large enough $n\in\mathbb{N}$, $x\notin K_n$.
That is, $x\notin\bigcap K_n$, so $\bigcap K_n$ is empty. But, Since each K_n is Compact,
thus $K_N=\emptyset$ for some $N\in\mathbb{N}$. That is,
$$0\leq g_N(x)<\varepsilon \text{ for all } x\in K,$$
and since $g_N\geq g_{N+1}\geq g_{N+1}\geq\cdots$, so $\sup|g_n - 0|\to 0$. ▨

「Thm. Let X be a Metric Space, and $\mathcal{C}(X)$ be the all Complex-Valued Conti, bounded functions on X.
Define $\|f\| = \sup_{x\in X}\|f(x)\|$. then it is norm on $\mathcal{C}(X)$. This norm Can derive a Metric on $\mathcal{C}(X)$,

Uniform Convergence and Integration.

Thm. Let $\alpha:[a,b]\to\mathbb{R}$ be monotone increasing and $f_n\in\mathcal{R}(\alpha)$, $n\in\mathbb{N}$.
If $f_n \rightrightarrows f$ on $[a,b]$, then $f\in\mathcal{R}(\alpha)$ and

$$\lim_{n\to\infty}\int_a^b f_n\, d\alpha = \int_a^b \lim_{n\to\infty} f_n\, d\alpha = \int_a^b f\, d\alpha.$$

Proof. For each $n\in\mathbb{N}$, define $\varepsilon_n=\sup_{x\in[a,b]}|f_n(x)-f(x)|$. Then, clearly

$$f_n-\varepsilon_n\le f\le f_n+\varepsilon_n \qquad (\text{for all } t\in[a,b])$$

Therefore, $$\int_a^b f_n-\varepsilon_n\, d\alpha \le \underline{\int_a^b} f\, d\alpha \le \overline{\int_a^b} f\, d\alpha \le \int_a^b f_n+\varepsilon_n\, d\alpha \qquad (*)$$

Hence $$0\le \overline{\int_a^b} f\, d\alpha - \underline{\int_a^b} f\, d\alpha \le 2\varepsilon_n\cdot[\alpha(b)-\alpha(a)]\to 0 \text{ as } n\to\infty.$$

Thus $f\in\mathcal{R}(\alpha)$. The equality is obtained by $(*)$, using $f\in\mathcal{R}(\alpha)$.

Uniform Convergence and Differentiation.

Thm. Let $f_n:[a,b]\to\mathbb{R}$ be differentiable maps s.t. for some $x_0\in[a,b]$, $\{f(x_0)\}$ converges.
If $\{f_n'\}$ converges uniformly on $[a,b]$, then $\{f_n\}$ converges uniformly on $[a,b]$ to f with

$$\frac{d}{dx}\lim_{n\to\infty} f_n(x)=\lim_{n\to\infty}\frac{d}{dx}f_n(x).$$

Proof. Given $\varepsilon>0$, take $N\in\mathbb{N}$ s.t. $n,m\ge N$ implies

$$|f_n(x_0)-f_m(x_0)|\le\frac{\varepsilon}{2} \text{ and } |f_n'(t)-f_m'(t)|\le\frac{\varepsilon}{2(b-a)},\ \forall t\in[a,b].$$

By the Mean-Value Theorem for f_n-f_m, for all $x,t\in[a,b]$,

$$\Big|f_n(x)-f_m(x)-\big(f_n(t)-f_m(t)\big)\Big|\le\frac{\varepsilon}{2(b-a)}\cdot(b-a)=\frac{\varepsilon}{2}. \qquad (*)$$

Therefore, for any $x,t\in[a,b]$ and $n,m\ge N$,

$$|f_n(x)-f_m(x)|\le|f_n(x)-f_m(x)-f_n(x_0)+f_m(x_0)|+|f_n(x_0)-f_m(x_0)|\le\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon.$$

so that $\{f_n\}$ converges uniformly, let $f=\lim_{n\to\infty} f_n$. For each $x\in[a,b]$, define on $[a,b]\setminus\{x\}$

$$\phi_n(t)=\frac{f_n(t)-f_n(x)}{t-x} \quad\text{and}\quad \phi(t)=\frac{f(t)-f(x)}{t-x}$$

$(*)$ gives ϕ_n is Cauchy sequence, thus converges uniformly, to ϕ. Finally, the fact that ϕ_n continuous gives the result.

「Summary. Let $f_n : X \to Y$ be functions, $f_n \to f$.

Continuity: If f_n are continuous and $f_n \rightrightarrows f$, then f is continuous.

Differentiation: If f_n are differentiable and $f_n' \rightrightarrows f'$ and $f_n(x_0) \to f(x_0)$, then $f_n \rightrightarrows f$ and

$$\frac{d}{dx} \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{d}{dx} f_n(x)$$

Integral: If $f_n \in \mathscr{R}(\alpha)$ and $f_n \rightrightarrows f$, then $f \in \mathscr{R}(\alpha)$ and

$$\int_a^b \lim_{n\to\infty} f_n \, d\alpha = \lim_{n\to\infty} \int_a^b f_n \, d\alpha$$

⌈Counter examples

⌈ $f_n : [0,1] \to \mathbb{R} ; x \mapsto x^n$ $\qquad$ f_n conti,

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1. \end{cases}$$ $\qquad$ f not conti

⌈ $f_n : [0,1] \to \mathbb{R} ; x \mapsto \frac{1}{n} x$ $\qquad$ f_n converges uniformly

$\tilde{f}_n : \mathbb{R} \to \mathbb{R} ; x \mapsto \frac{1}{n} x$ $\qquad$ $\tilde{f}_n$ not converges uniformly

⌈ $f_n : [0,2] \to \mathbb{R} ; x \mapsto \dfrac{1-x}{1+x^n}$ $\qquad$ f_n differentiable, converges uniformly, but

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 1-x & \text{if } 0 \le x \le 1 \\ 0 & \text{if } 1 < x \le 2. \end{cases}$$ $\qquad$ f not differentiable at $x=1$

In $[0,1]$, $\displaystyle |f_n(x) - f(x)| = \left| (1-x)\cdot\left(\frac{1}{1+x^n} - 1\right) \right| = \frac{|1-x|\cdot|x^n|}{|1+x^n|} \le |1-x|\cdot|x^n| = (1-x)\cdot x^n.$

Since $((1-x)\cdot x^n)' = -x^n + (1-x) n \cdot x^{n-1} = n x^{n-1} - (n+1)x^n = x^{n-1}(n-(n+1)x)$

has a maximum at $x = \frac{n}{n+1}$, $\displaystyle \left(1 - \frac{n}{n+1}\right)\cdot\left(\frac{n}{n+1}\right)^n = \frac{n^n}{(n+1)^{n+1}} \to 0$ as $n \to \infty$.

So, $\displaystyle \sup_{[0,1]} |f_n(x) - f(x)| \to 0$ as $n \to \infty$.

In $[1,2]$, $\displaystyle |f_n(x) - f(x)| = \left| \frac{1-x}{1+x^n} \right| = \frac{x-1}{1+x^n} \le \frac{x-1}{x^n} = \frac{1}{x^{n-1}} - \frac{1}{x^n} \to 0$ as $n \to \infty$.

Now, $f_n \to f$ converges uniformly.

⌈ $f_n : \mathbb{R} \to \mathbb{R} ; x \mapsto \frac{1}{\sqrt{n}} \cdot \sin(nx)$. $\qquad$ f_n differentiable, converges uniformly,

$f \equiv 0$, since $|\sin(nx)| \le 1$ $\qquad$ but

$f_n' : \mathbb{R} \to \mathbb{R} ; x \mapsto \sqrt{n} \cdot \cos(nx)$ $\qquad$ f_n' not converges pointwise.

Γ Let $X_n = \{x_n \mid n \in \mathbb{N}\} = \mathbb{Q} \cap [0,1]$.

$f_n : [0,1] \to \mathbb{R} : x \mapsto \begin{cases} 1 & \text{if } x \in X_n \\ 0 & \text{if } x \notin X_n. \end{cases}$ f_n Riemann-integralable, since it has finite disconti

but

$f : [0,1] \to \mathbb{R} : x \mapsto \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$ f not integrable, it is Diriclet.

Γ

$f_n : [0,1] \to \mathbb{R} ; x \mapsto \begin{cases} n & \text{if } 0 < x < \frac{1}{n} \\ 0 & \text{if } x = 0 \text{ on } \frac{1}{n} \leq x \leq 1. \end{cases}$

$\lim_{n\to\infty} f_n \equiv 0$, but $\int_0^1 f_n = 1$. Cannot change limit under the integral.

Γ $f_n : \mathbb{R} \to \mathbb{R} ; x \mapsto \dfrac{x^2}{(1+x^2)^n}$ f_n continuous, but the sum is not continuous.

$f(x) = \sum_{n=0}^{\infty} f_n(x) = \sum_{n=0}^{\infty} \dfrac{x^2}{(1+x^2)^n} = \begin{cases} 0 & \text{if } x = 0 \\ 1+x^2 & \text{if } x \neq 0 \end{cases}$ $\left(x^2 \cdot \dfrac{1}{1 - \frac{1}{1+x^2}} \right)$

Γ $f_n : [0,1] \to \mathbb{R} ; x \mapsto nx \cdot (1-x^2)^n$ f_n differentiable,

$\lim_{n\to\infty} f_n(x) = 0$

$\int_0^1 x(1-x^2)^n dx = \int_1^0 -\frac{1}{2} \cdot t^n dt = \frac{1}{2} \cdot \frac{1}{n+1} t^{n+1} \Big|_0^1 = \frac{1}{2} \cdot \frac{1}{n+1}$.

So, $\int_0^1 f_n(x) dx = \frac{1}{2} \cdot \frac{n}{n+1} \to \frac{1}{2}$, but $\int_0^1 f \, dx = 0$,

Note; replace nx to $n^2 x$, then $\int_0^1 f_n \to \infty$.